**Statistical Model to Predict Price of Amazon Equity**

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## I. Abstract

Using various predictive analytics tools in R, the aim of the study is to build a model to analyze the stock price of Amazon using the variables earnings per share (EPS) and book value per share (BVPS). The data of interest are companies in the insurance industry from the S&P 1500, which contains the 1,500 largest companies with 11 unique sectors and 77 industries. Amazon accounts for 1.11% of total sales (revenue) among S&P 1500 companies, totaling $2,710,664 out of $243,368,093.

## II. Introduction

This analysis uses annual financial data from FactSet, reported at one-year intervals. This study will build a model to analyze the stock price of Amazon. Predictive analytics tools will use multiple linear, parametric regression. Investors, investment bankers, and portfolio managers would be interested in the results of the study.

## III. Previous Research

There is no previous research focusing on this topic.

## IV. Methodology

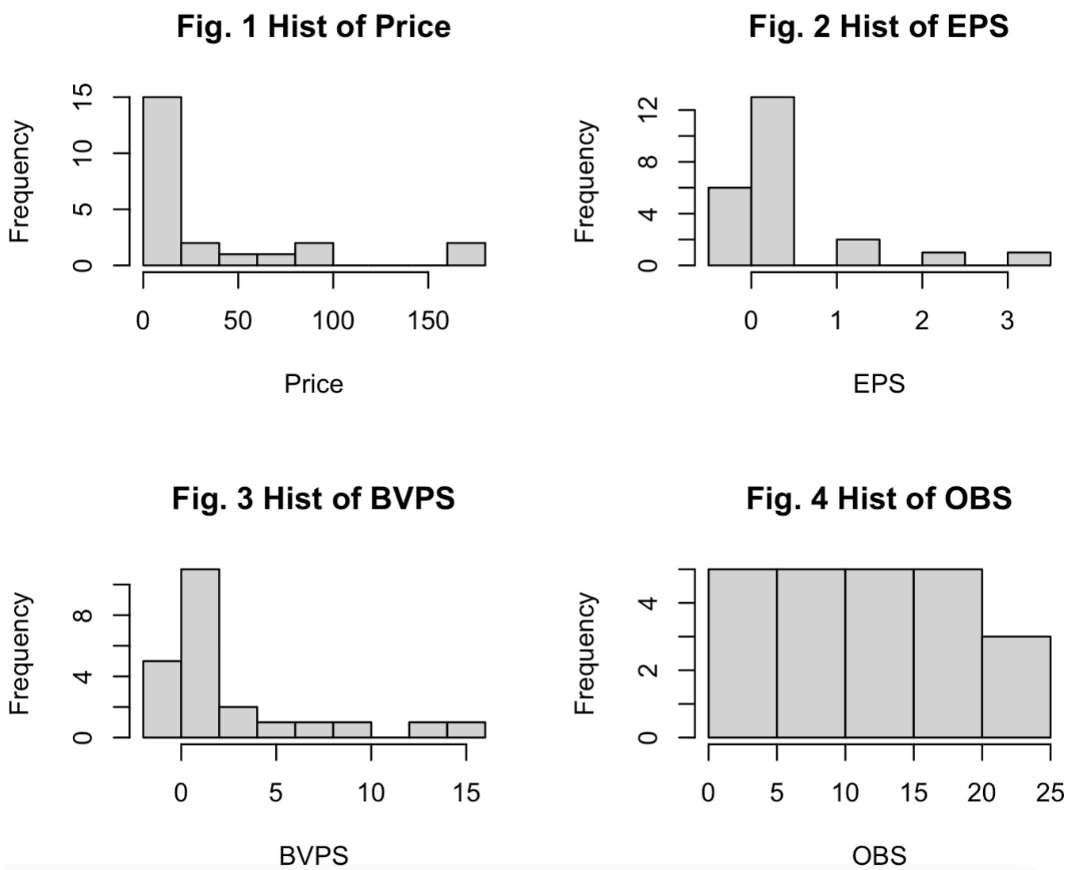
This study is a time series analysis that uses data from the S&P 1500 from Factset. The data ranges from 2000 to 2022, and focuses on the company Amazon. All the data processing was done using R programming. This analysis uses histograms, time series plots, and scatter plots, as well as various analytical techniques, such as descriptive statistics, correlation matrix, and regression results for both linear regression and robust linear regression.

Eqn 1 (Functional specification) Price ~ EPS (+) , BVPS (+), OBS (+)

Eqn 2 (Population Regression) $Price_i = \alpha + \beta_1 EPS + \beta_2 BVPS + \beta_3 OBS$

Eqn 3 (Sample regression) $Price_i = a + b_1 EPS + b_2 BVPS + b_3 OBS$

## V. Results




Figures 1 through 4 display the histograms of the frequency of each variable: Price, EPS, BVPS, and OBS. All of the figures, except Figure 4 are skewed to the right. Additionally, Figure 1 displays potential outliers.

Fig. 5 Timeseries of Price

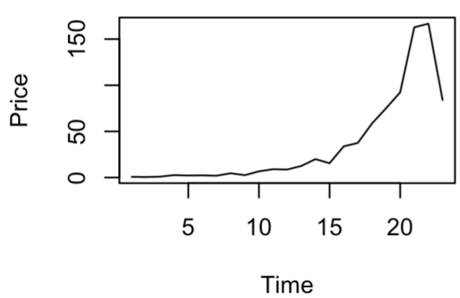



Fig. 6 Timeseries of EPS

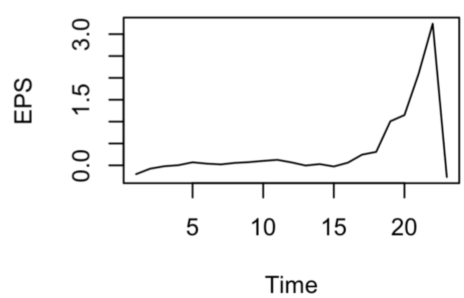



Fig. 7 Timeseries of BVPS

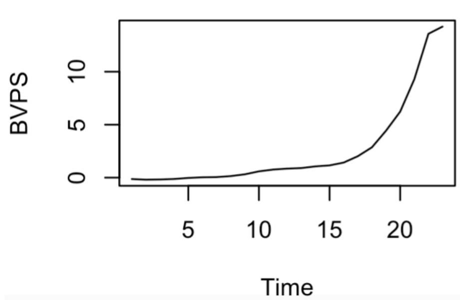
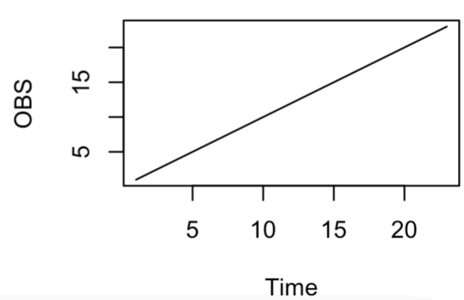



Fig. 8 Timeseries of OBS




Figures 5 through 8 display the time series plots of each variable: Price, EPS, BVPS, and OBS. Figures 5 through 8 demonstrate that the variables are trending up. However, there may be an outlier in Figures 5 and 6 causing the sudden drop.

**Fig. 9 Scatterplot of Price v EPS**

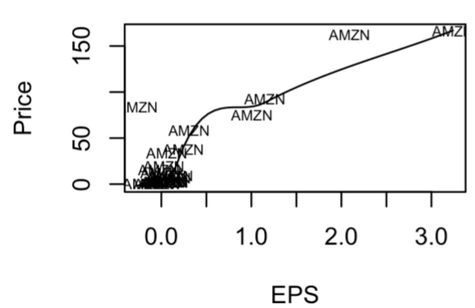



**Fig. 10 Scatterplot of Price v BVPS**

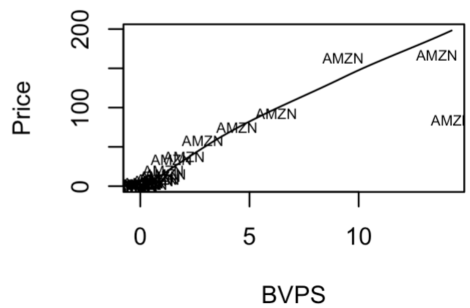
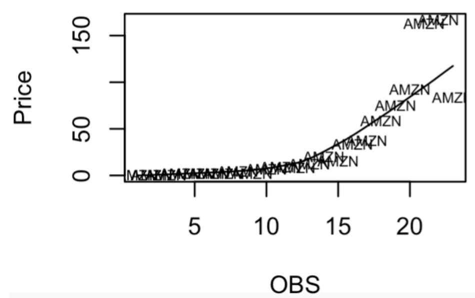



**Fig. 11 Scatterplot of Price v OBS**




Figures 9 through 11 display the scatterplots of the relationship of each variable (EPS, BVPS, and OBS) with price. All of the figures are relatively strong. All of the figures are positive and Figure 9 indicates that the relationship between price and EPS may be nonlinear.

**Table 1**
*Descriptive Statistics*

| Name | Obs | Max | Min | Mean | Median | Std | Skew | Kurt |
|---|---|---|---|---|---|---|---|---|
| Price | 23 | 166.72 | 0.54 | 34.86 | 9.00 | 49.82 | 1.54 | 4.65 |
| EPS | 23 | 3.24 | -0.27 | 0.35 | 0.06 | 0.82 | 2.33 | 8.50 |
| BVPS | 23 | 14.26 | -0.19 | 2.58 | 0.85 | 4.26 | 1.74 | 5.19 |

Table 1 demonstrates the descriptive statistics.

**Table 2**
*Correlation Matrix*

| | Price | EPS | BVPS |
|---|---|---|---|
| Price | 1.000 | 0.883 | 0.895 |
| EPS | 0.883 | 1.000 | 0.681 |

| | | | |
|---|---|---|---|
| BVPS | 0.895 | 0.681 | 1.000 |

Table 2 demonstrates the correlation between the different variables. The variables EPS and BVPS have a strong, positive correlation with price. There is a slight collinearity between EPS and BVPS (0.681).

**Table 3**
*Regression Results*

| | Estimate | Stnd. Error | T Value | Pr(>\|t\|) |
|---|---|---|---|---|
| (Intercept) | -6.481 | 5.633 | -1.151 | 0.264201 |
| EPS | 30.253 | 3.954 | 7.650 | 3.23e-07 |
| BVPS | 4.536 | 1.007 | 4.506 | 0.00242 |
| OBS | 1.583 | 0.563 | 2.812 | 0.011134 |

*Residual standard error: 11.06 on 19 degrees of freedom
*Multiple R-squared: 0.9574, Adjusted R-squared: 0.9507
*F-statistic: 142.4 on 3 and 19 DF, p-value: 3.361e-13

Eqn 4 (Regression line) Price = -6.481+ 30.253 EPS + 4.536 BVPS + 1.583 OBS

Table 3 demonstrates the regression coefficients and has Eqn 4 or the regression line. All of the variables, EPS, BVPS, and OBS have significant coefficients. Additionally, 95.74% of the total variation of price is explained by the variation of the independent variables. The model is also significant. As EPS goes up by a dollar, price goes up $30.25 on average, other things equal. As BVPS goes up by a dollar, price goes up $4.53 on average, other things equal.

Fig. 12 Hist of Residuals

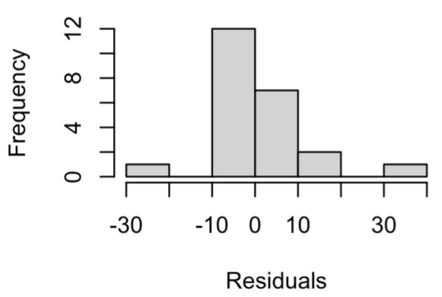



Fig. 13 Act v Pred Price

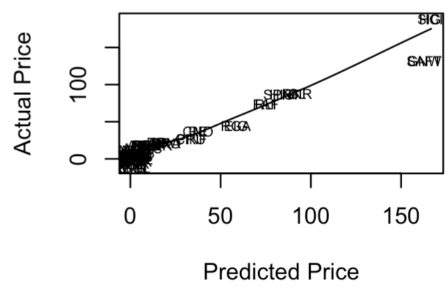
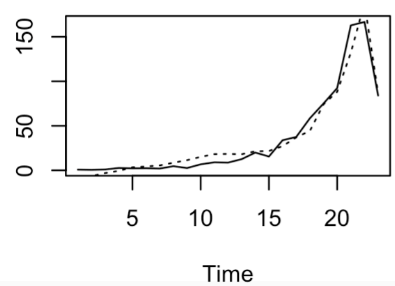



Fig. 14 Timeseries of Price




Figures 12 through 14 display the residuals. Figure 12 indicates that the histogram of the residuals is slightly normally distributed. Figure 13 shows the scatterplot of the actual against the predicted price, which is strong and positive. Lastly, Figure 14 demonstrates that the time series plot of actual against predicted price track each other well.

**Table 4**
*Regression Results (With Robust)*

| | Estimate | Stnd. Error | T Value | Pr(>\|t\|) |
|---|---|---|---|---|
| (Intercept) | 4.9000 | 2.6329 | 1.861 | 0.0783 |
| EPS | -1.0177 | 12.2379 | -0.083 | 0.9346 |
| BVPS | 18.6811 | 3.2412 | 5.764 | 1.49e-05 |
| OBS | -0.6178 | 0.3577 | -1.727 | 0.1004 |

*Residual standard error: 3.815 on 19 degrees of freedom
*Multiple R-squared: 0.4725

Eqn 5 (Regression line with Robust) Price = 4.900 - 1.01 EPS + 18.68 BVPS - 0.61 OBS
Table 4 demonstrates the regression coefficients and has Eqn 4 or the regression line using the Robust library. All of the variables, BVPS and OBS have significant coefficients. Additionally, 47.25% of the total variation of price is explained by the variation of the independent variables. The model is also significant. As BVPS goes up by a dollar, price goes up $18.68 on average, other things equal.

Fig. 15 Hist of Residuals (With Robust)

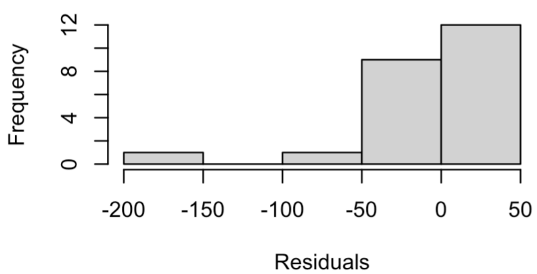



Fig. 16 Act v Pred Price (With Robust)

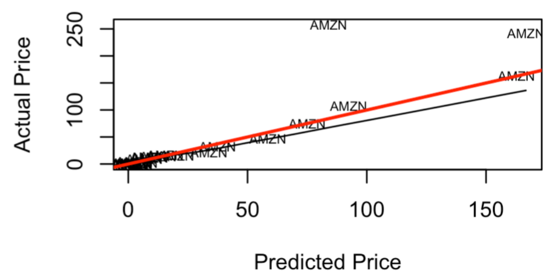
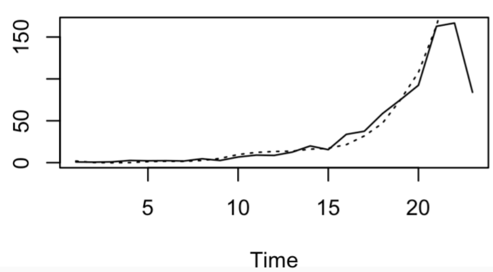



Fig. 17 Timeseries of Price (With Robust)




Figures 15 through 17 display the residuals after implementing the Robust library. Figure 15 shows that the histogram of the residuals is skewed to the left. Figure 16 shows the scatterplot of the actual against the predicted price, which is strong and positive. Lastly, Figure 17 demonstrates that the time series plot of actual against predicted price track each other well.

## VI. <u>Conclusion</u>

In conclusion, the study was successful. The OLS model was significant (F-statistic = 142.4) with an $R^2$ of 95.74%, suggesting excellent fit. The robust regression, though with a lower $R^2$ of 47.25%, accounts for potential outliers, emphasizing BVPS as the most consistent predictor. Future research might expand by analyzing more variables.

## VII. <u>Bibliography</u>

Data obtained from: Dr. Russon's course materials; Factset

## VIII. <u>Appendix I</u>

```
      tkr             name   price    eps   bvps      sales year obs
1959 AMZN Amazon.com, Inc.   0.778 -0.201 -0.135   2761.983 2000   1
1964 AMZN Amazon.com, Inc.   0.541 -0.076 -0.193   3122.433 2001   2
1972 AMZN Amazon.com, Inc.   0.944 -0.020 -0.174   3932.936 2002   3
1961 AMZN Amazon.com, Inc.   2.631  0.004 -0.128   5263.699 2003   4
1963 AMZN Amazon.com, Inc.   2.215  0.070 -0.028   6921.124 2004   5
1960 AMZN Amazon.com, Inc.   2.357  0.039  0.030   8490.000 2005   6
1967 AMZN Amazon.com, Inc.   1.973  0.022  0.052  10711.000 2006   7
1971 AMZN Amazon.com, Inc.   4.632  0.056  0.144  14835.000 2007   8
1956 AMZN Amazon.com, Inc.   2.564  0.074  0.312  19166.000 2008   9
1969 AMZN Amazon.com, Inc.   6.726  0.102  0.592  24509.000 2009  10
1962 AMZN Amazon.com, Inc.   9.000  0.126  0.761  34204.000 2010  11
1966 AMZN Amazon.com, Inc.   8.655  0.068  0.852  48077.000 2011  12
1977 AMZN Amazon.com, Inc.  12.544 -0.004  0.902  61093.000 2012  13
1968 AMZN Amazon.com, Inc.  19.939  0.029  1.062  74452.000 2013  14
1976 AMZN Amazon.com, Inc.  15.518 -0.026  1.155  88988.000 2014  15
1958 AMZN Amazon.com, Inc.  33.795  0.062  1.421 107006.000 2015  16
1957 AMZN Amazon.com, Inc.  37.493  0.245  2.021 135987.000 2016  17
1975 AMZN Amazon.com, Inc.  58.474  0.308  2.862 177866.000 2017  18
1978 AMZN Amazon.com, Inc.  75.098  1.007  4.435 232887.000 2018  19
1970 AMZN Amazon.com, Inc.  92.392  1.150  6.231 280522.000 2019  20
1965 AMZN Amazon.com, Inc. 162.846  2.091  9.285 386064.000 2020  21
1973 AMZN Amazon.com, Inc. 166.717  3.239 13.580 469822.000 2021  22
1974 AMZN Amazon.com, Inc.  84.000 -0.267 14.259 513983.000 2022  23
```

*Original data (spInfo and spData) too large to paste, so this is the tsdf (time series dataframe), which was used for all analysis.

## IX. <u>Appendix II</u>

```
# Leslie Alhakim
# BUA 633 Term Project 2

# Import packages
library(YRmisc)
library(robust)

# Import datasets
library(readxl)
spInfo <- read_excel("Downloads/BUA 633/spInfo-1 (1).xlsx")
View(spInfo)

spData <- read_excel("Downloads/BUA 633/spData-1 (1).xlsx")
View(spData)

# Preliminaries
names(spData)
names(spInfo)

data.class(spData)
spData<-as.data.frame(spData)
data.class(spData)
```

```
data.class(spInfo)
spInfo<-as.data.frame(spInfo)
data.class(spInfo)

# Merge dataframes
spdf<-merge(spData,spInfo,by="tkr")
dim(spdf)

names(spdf)

# Extract "year" variable
spdf$date<-as.numeric(substring(spdf$date,7,10))
names(spdf)[2]<-"year"
names(spdf)

# Choose a company tkr: AMAZON
unique(spdf$tkr)
names(spdf)
dim(spdf)
tsdf<-spdf[spdf$tkr=="AMZN",c("tkr",
"name","price","eps","bvps",
                              "sales","year")]
tsdf
tsdf<-df.sortcol(tsdf,"year",FALSE)
dim(tsdf)
names(tsdf)
tsdf$obs<-1:23
names(tsdf)
tsdf[,3:6]<-round(tsdf[,3:6],3)

# Graphical techniques: HISTOGRAMS
histogram <-function(x,myTitle,xxlab,yylab){
  hist(x,main=myTitle,xlab=xxlab,ylab=yylab)}

par(mfrow=c(2,2))
histogram(tsdf$price,"Fig. 1 Hist of Price","Price","Frequency")
histogram(tsdf$eps,"Fig. 2 Hist of EPS", "EPS","Frequency")
histogram(tsdf$bvps,"Fig. 3 Hist of BVPS", "BVPS","Frequency")
histogram(tsdf$obs,"Fig. 4 Hist of OBS", "OBS","Frequency")

# Graphical techniques: TIMESERIES PLOTS
plotts<- function(x,myTitle,yylab){
  ts.plot(x,main=myTitle,ylab=yylab)}

par(mfrow=c(2,2))
plotts(tsdf$price, "Fig. 5 Timeseries of Price", "Price")
```

```
plotts(tsdf$eps, "Fig. 6 Timeseries of EPS", "EPS")
plotts(tsdf$bvps, "Fig. 7 Timeseries of BVPS", "BVPS")
plotts(tsdf$obs, "Fig. 8 Timeseries of OBS", "OBS")

# Graphical techniques: SCATTERPLOTS
plotTkr<-function(x,y,z,xxlab,yylab,myTitle){
 scatter.smooth(x,y,type="n",xlab=xxlab,ylab=yylab,main=myTitle)
  text(x,y,z,cex=.6)   }

par(mfrow=c(2,2))
plotTkr(tsdf$eps,tsdf$price,tsdf$tkr,"EPS","Price","Fig. 9
Scatterplot of Price v EPS")
plotTkr(tsdf$bvps,tsdf$price,tsdf$tkr,"BVPS","Price","Fig. 10
Scatterplot of Price v BVPS")
plotTkr(tsdf$obs,tsdf$price,tsdf$tkr,"OBS","Price","Fig. 11
Scatterplot of Price v OBS")

# Analytical methods: DESCRIPTIVE STATISTICS
ds.summ(tsdf[,c("price","eps","bvps","obs")],2)

# Analytical methods: CORRELATION
round(cor(na.omit(tsdf[,c("price","eps","bvps")])),3)

# Analytical methods: REGRESSION RESULTS
fit<-lm(price ~ eps+bvps+obs,na.action=na.omit,data=tsdf)
summary(fit)

# Post regression validation
par(mfrow=c(2,2))
histogram(fit$residuals,"Fig. 12 Hist of Residuals",
"Residuals","Frequency")
plotTkr(fit$model$price,fit$fitted.values,tsdf$tkr, "Predicted
Price","Actual Price","Fig. 13 Act v Pred Price")
pl.2ts(tsdf$price,fit$fitted.values,"Fig. 14 Timeseries of
Price")

# Graphical methods: ROBUST
fit<-lmRob(price ~ eps+bvps+obs,na.action=na.omit,data=tsdf)
cor(tsdf$price,predict(fit,tsdf))^2
summary(fit)

par(mfrow=c(2,2))
histogram(fit$residuals,"Fig. 15 Hist of Residuals (With
Robust)", "Residuals","Frequency")
plotTkr(fit$model$price,fit$fitted.values,tsdf$tkr, "Predicted
Price","Actual Price","Fig. 16 Act v Pred Price (With Robust)")
abline(a = 0, b = 1, col = "red", lwd = 2)
```

```
pl.2ts(tsdf$price,fit$fitted.values,"Fig. 17 Timeseries of Price
(With Robust)")
```